Spacecraft ADCS Homework 3 Report
NASA Starling Mission Analysis

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Changelog

Date	Section Modified	Description of Changes
2026-02-25	HW1	Initial report completed
2026-03-7		Research
2026-03-09	Section 1/2	
2026-03-10	Section 3 + 1/2	Matrices
2026-03-12	Section 4	Solvers ran & results compiled
2026-03-13	Sections 6-9	Report 2 Complete
2026-03-30	HW3 Research	Research for sensor models

Project Repository

A GitHub repository has been created to store simulation code, numerical tools, and documentation for this project.

Repository Link:
`https://github.com/ATG218/spacecraft-adcs-course`

All simulations are implemented in **Julia**, running in Jupyter notebooks (`.ipynb`). Orbital dynamics are numerically integrated using the `DifferentialEquations.jl` package. 3D orbit animations are produced using `MeshCat.jl`.

1 Course Project Logistics

For this project, we selected NASA's **Starling** mission as the spacecraft platform for analysis. Each homework assignment will expand upon this document and ultimately be compiled into a final report on our work.

2 Mission Description and ADCS Requirements

2.1 Mission Overview

NASA's **Starling** mission consists of four identical 6U CubeSats operating in low Earth orbit. The mission demonstrates autonomous swarm technologies including:

- Autonomous relative maneuver planning
- Optical-based relative navigation
- Inter-satellite networking
- Distributed onboard autonomy

The spacecraft were launched on July 17, 2023 and operate in a Sun-synchronous orbit at approximately 480 km altitude.

Although Starling is a multi-spacecraft mission, this will project analyze a single spacecraft as an isolated rigid body.

2.2 Spacecraft Bus

Each Starling spacecraft is built on a Blue Canyon Technologies (BCT) XB1 6U CubeSat bus. [9]

The standard 6U CubeSat dimensions are approximately:

$$0.10 \text{ m} \times 0.20 \text{ m} \times 0.30 \text{ m}$$

The bus integrates:

- Onboard avionics and computing
- Power generation and storage
- GNSS receiver
- Communications subsystem
- Attitude Determination and Control System (ADCS)

2.3 Attitude Determination and Control System

The Blue Canyon XB1 bus includes an integrated three-axis ADCS.

The ADCS system includes:

- Star trackers for fine attitude determination
- Inertial measurement unit (gyroscopes)

- Reaction wheels for three-axis control
- Magnetorquers for momentum dumping

Blue Canyon 6U systems report pointing accuracy on the order of 0.002 degrees (1σ) for dual-star-tracker configurations. The XACT family is rated for a slew rate of $\geq 10\,^\circ/\text{s}$ for a 4 kg, 3U CubeSat [10]. Scaled to the Starling 6U bus mass of approximately 12 kg, the effective slew rate is estimated at 3-5 $^\circ/\text{s}$, consistent with the higher inertia of the larger spacecraft.

2.4 ADCS Requirements

Based on mission objectives and spacecraft architecture, the following capabilities are required [6]:

- Three-axis stabilized control
- Reaction wheel authority to counter environmental torques
- Momentum dumping via magnetorquers
- Controlled slewing between pointing targets

Further quantitative analysis will follow in subsequent sections.

3 Spacecraft Model

3.1 Body Reference Frame Definition

A body-fixed reference frame $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ is defined and attached to the spacecraft. The origin of $\mathcal{B}$ is located at the spacecraft center of mass.

The axes of $\mathcal{B}$ are aligned with the principal geometric dimensions of the 6U spacecraft bus:

$$L_1 = 0.10 \text{ m}, \quad L_2 = 0.20 \text{ m}, \quad L_3 = 0.30 \text{ m}$$

The body axes are defined as follows:

- $\mathbf{b}_1$: aligned with the Y-direction, shortest dimension (0.10 m)
- $\mathbf{b}_2$: aligned with the X-direction, intermediate dimension (0.20 m)
- $\mathbf{b}_3$: aligned with the Z-direction, longest dimension (0.30 m)

The frame is defined to be right-handed such that

$$\mathbf{b}_1 \times \mathbf{b}_2 = \mathbf{b}_3.$$

The deployed solar panels are mounted on the $\pm\mathbf{b}_1$ faces of the spacecraft. The hinge runs along the $\mathbf{b}_2$ direction (20 cm), and when deployed the panels extend outward in the $\pm\mathbf{b}_1$ direction, with one lying approximately in the $\mathbf{b}_1$–$\mathbf{b}_2$ plane, and one tilted from that plane by $45\,^\circ$ with the $-\mathbf{b}_1$ axis.

This body-frame definition aligns the coordinate system with the dominant geometric features of the spacecraft and simplifies subsequent inertia and surface discretization calculations.

3.2 Modeling, Center of Mass, and COM Inertia

To gather the COM and the Inertial Matrix for our chosen spacecraft, we first created a model in Fusion360 (Figure 1). This model was a reduced, superficial version of the real Starling spacecraft. We broke the vehicle down into the main body frame (a solid rectangle of uniform density) and the main deployable solar arrays (solid thin rectangles, also of uniform density).

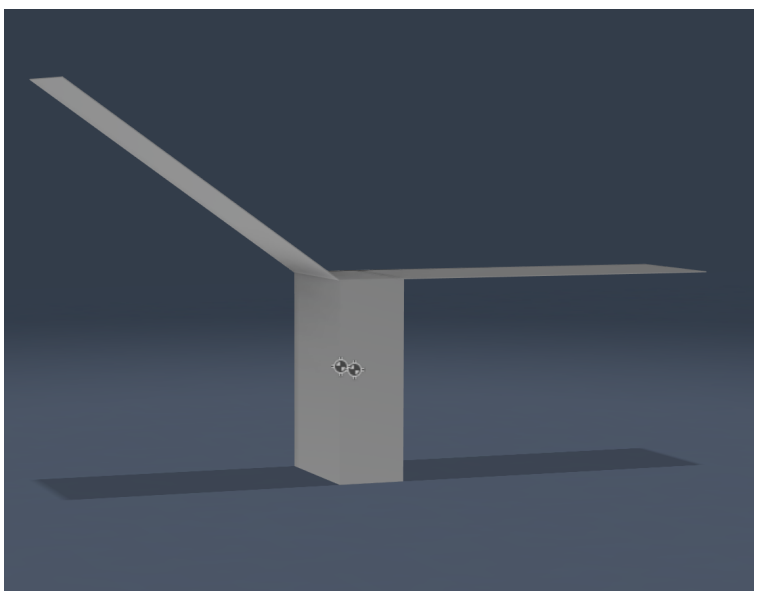

Figure 1: Basic Fusion 360 CAD model of the Starling satellite

Using our previously established density for the main body and an estimate of the density of the solar arrays gathered from standard CubeSat dimensions (1.5 mm thickness and areal density of about 5 kg/m^2) we were able to obtain the Center of Mass: thin

$$x_{COM} = 0.0 \text{ m}, \quad y_{COM} = 0.002 \text{ m}, \quad z_{COM} = 0.167 \text{ m}$$

And the Inertial Matrix:

$$\mathbf{I} = \begin{bmatrix} 0.249 & 0.00 & 0.00 \\ 0.00 & 0.187 & 0.02 \\ 0.00 & 0.02 & 0.148 \end{bmatrix}$$

Should probably not be diagonal

3.3 Principal Axis and Rotation of Body

The principal moments of inertia and the rotation matrix from the body frame to the principal frame were computed numerically using Julia's built-in `eigen()` function applied to the inertia matrix from Section 3.2, giving:

$$I_1 = 0.13957 \text{ kg m}^2 \quad \text{(minor)},$$
$$I_2 = 0.19543 \text{ kg m}^2 \quad \text{(intermediate)},$$
$$I_3 = 0.249 \text{ kg m}^2 \quad \text{(major)}.$$

These principal moments are used directly in the Euler's equation simulation (Section 5).

Doesn't match the above

3.4 Surface Discretization

With our simple model, we are able to discretize the outer surfaces of the satellite into 10 distinct surfaces, 6 from the body itself, and 4 from the solar arrays. For now, we will solely consider the deployed state of the solar panels, in which case they can be discretized simply.

Table 1: Surface Discretization of Satellite Exterior

Surface	**Centroid** (x, y, z) **[m]**	**Normal Vector** $\hat{n}$	**Area [m^2]**	**Description**
Bus +X	$(0.1,\ 0,\ 0.15)$	$(+1,\ 0,\ 0)$	0.03 m^2	Parallel to b2
Bus -X	$(-0.1,\ 0,\ 0.15)$	$(-1,\ 0,\ 0)$	0.03 m^2	Parallel to b2
Bus +Y	$(0,\ 0.05,\ 0.15)$	$(0,\ +1,\ 0)$	0.06 m^2	Parallel to b1
Bus -Y	$(0,\ -0.05,\ 0.15)$	$(0,\ -1,\ 0)$	0.06 m^2	Parallel to b1
Bus +Z	$(0,\ 0,\ 0.3)$	$(0,\ 0,\ +1)$	0.02 m^2	Parallel to b3
Bus -Z	$(0,\ 0,\ 0)$	$(0,\ 0,\ -1)$	0.02 m^2	Parallel to b3
Panel Right (Outer)	$(0,\ 0.3,\ 0.3015)$	$(0,\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}})$	0.10 m^2	Perpendicular to b1
Panel Right (Inner)	$(0,\ 0.3,\ 0.30)$	$(0,\ -\frac{1}{\sqrt{2}},\ -\frac{1}{\sqrt{2}})$	0.10 m^2	...
Panel Left (Outer)	$(0,\ 0.254,\ 0.4445)$	$(0,\ 0,\ +1)$	0.10 m^2	35.3 to b1
Panel Left (Inner)	$(0,\ 0.255,\ 0.443)$	$(0,\ 0,\ -1)$	0.10 m^2	...

4 Dynamics

4.1 State Vector and Reference Frame

The spacecraft position and velocity are expressed in an Earth-Centered Inertial (ECI) frame. We used seconds and kilometers throughout, as recommended in the instructions.

The 6-element state vector used is

$$\mathbf{x} = \begin{bmatrix} \mathbf{r} \\ \mathbf{v} \end{bmatrix} \in \mathbb{R}^6,$$

where $\mathbf{r} = [x,\, y,\, z]^\top$ (km) is the ECI position and $\mathbf{v} = [\dot{x},\, \dot{y},\, \dot{z}]^\top$ (km/s) is the ECI velocity.

4.2 Equations of Motion

The orbital ODE we used is

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{a}_{2B} + \mathbf{a}_{J_2} + \mathbf{a}_D.$$

The following are all of the considerations made to the dynamics of our system:

Two-Body Gravity

$$\mathbf{a}_{2B} = -\frac{\mu}{r^3}\,\mathbf{r}, \qquad r = \|\mathbf{r}\|,$$

with $\mu = 398600.4418\ \mathrm{km}^3/\mathrm{s}^2$.

J_2, Oblateness Perturbation

Earth's oblateness causes the orbital plane to slowly rotate in inertial space which results in a secular drift of the right ascension of the ascending node (RAAN). Physically, the asymmetric mass distribution generates a small torque on the satellite's orbital angular momentum vector, changing its direction over time while preserving its magnitude. This effect is significant at low orbits and must be accounted for when modeling long-term orbital evolution.

In Earth-centered inertial (ECI) coordinates, the acceleration due to the J_2 perturbation is given by [11]

$$\mathbf{a}_{J_2} = \frac{3\,J_2\,\mu\,R_E^2}{2\,r^5} \begin{bmatrix} x\left(\dfrac{5z^2}{r^2} - 1\right) \\ y\left(\dfrac{5z^2}{r^2} - 1\right) \\ z\left(\dfrac{5z^2}{r^2} - 3\right) \end{bmatrix},$$

where μ is Earth's gravitational parameter, R_E is Earth's equatorial radius, and (x, y, z) is the spacecraft position in the inertial frame. The parameter $J_2 = 1.08262668 \times 10^{-3}$ quantifies the magnitude of Earth's oblateness, and $R_E = 6378.1363$ km is the reference equatorial radius, both consistent with the EGM96 gravity model [15].

Atmospheric Drag

An exponential atmosphere model is used with reference density $\rho_0 = 3.614 \times 10^{-13}$ kg/m^3 at reference altitude $h_0 = 700$ km and scale height $H = 88.667$ km [14]:

$$\rho(h) = \rho_0 \exp\left(-\frac{h - h_0}{H}\right), \qquad h = \|\mathbf{r}\| - R_E.$$

The drag acceleration is

$$\mathbf{a}_D = -\frac{1}{2}\,\rho\, B_c\, |\mathbf{v}_{\text{rel}}|\, \mathbf{v}_{\text{rel}},$$

where $\mathbf{v}_{\text{rel}} = \mathbf{v} - \boldsymbol{\omega}_E \times \mathbf{r}$ accounts for Earth's rotation ($\omega_E = 7.2921159 \times 10^{-5}$ rad/s), and $B_c = C_d A/m$ is the ballistic coefficient. Representative values $C_d = 2.2$, $A = 0.03$ m^2, and $m = 12$ kg give us a ballistic coefficient $B_c = 5.50 \times 10^{-3}$ m^2/kg. The cross-sectional area presented to the flow varies with spacecraft attitude, which is not modeled in the current simulation as we do not take into account attitude determination.

4.3 Sun-Synchronous Orbit Design

A sun-synchronous orbit (SSO) requires the orbital plane to precess at Earth's mean heliocentric rate of $\dot{\Omega} = +0.985647\,^\circ$/day. The secular RAAN drift due to J_2 for a near-circular orbit is [11]

$$\dot{\Omega} = -\frac{3}{2}\, J_2\, n \left(\frac{R_E}{p}\right)^2 \cos i, \qquad n = \sqrt{\frac{\mu}{a^3}}, \quad p = a(1 - e^2).$$

Setting $\dot{\Omega} = +0.985647\,^\circ$/day and $e \approx 0$, the required inclination satisfies

$$\cos i = \frac{\dot{\Omega}}{-\frac{3}{2}\, J_2\, n\, (R_E/a)^2}.$$

For the Starling altitude of 480 km, $a = 6858.136$ km, and solving gives $i = 97.326^\circ$, a retrograde inclination as expected for SSO.

4.4 Initial Conditions

Initial conditions are specified via classical orbital elements (COEs) and converted to ECI coordinates using the standard perifocal-to-ECI direction cosine matrix $Q = R_3(\Omega)\, R_1(i)\, R_3(\omega)$ [11]. The epoch state is:

Semi-major axis	$a = 6858.136$ km
Eccentricity	$e = 0$
Inclination	$i = 97.326^\circ$
RAAN	$\Omega = 0^\circ$
Arg. of perigee	$\omega = 0^\circ$
True anomaly	$\nu = 0^\circ$

This yields the initial ECI state $\mathbf{r}_0 = [6858.136,\ 0,\ 0]^\top$ km and $\mathbf{v}_0 = [0,\ -0.9722,\ 7.5615]^\top$ km/s, with orbital speed $|\mathbf{v}_0| = 7.6237$ km/s and orbital period $T = 5652$ s (94.2 min).

4.5 Numerical Simulation and Results

The ODE is integrated using the Vern9 (9th-order Runge–Kutta) solver from the Julia `DifferentialEquations.jl` package, with relative tolerance 10^{-10} and absolute tolerance 10^{-12}. The simulation spans 10 orbital periods (≈ 15.7 hours), with output saved every 30 s. Two cases are run:

1. **J2-only (no drag).** The RAAN precession rate extracted by a least-squares linear fit over 10 orbits is $\dot{\Omega} = 0.991\,^\circ$/day, compared to the target $0.985647\,^\circ$/day (this is an error of 0.54%). The small discrepancy is due to the secular formula being a first-order orbital average, while the simulation propagates osculating elements that include short-period J_2 oscillations. The semi-major axis is conserved to within 43 m over 15.7 hours, confirming that J_2 produces no secular energy change for a circular orbit.

2. **J2 + drag.** Over 15.7 hours, atmospheric drag reduces the semi-major axis by $\Delta a = -0.119$ km.

Perturbations not included in the current model are solar radiation pressure, third-body gravity, and spacecraft maneuvers Figures 2, 3, and 4 show the simulated ECI trajectory, RAAN evolution, and ground track with altitude decay respectively.

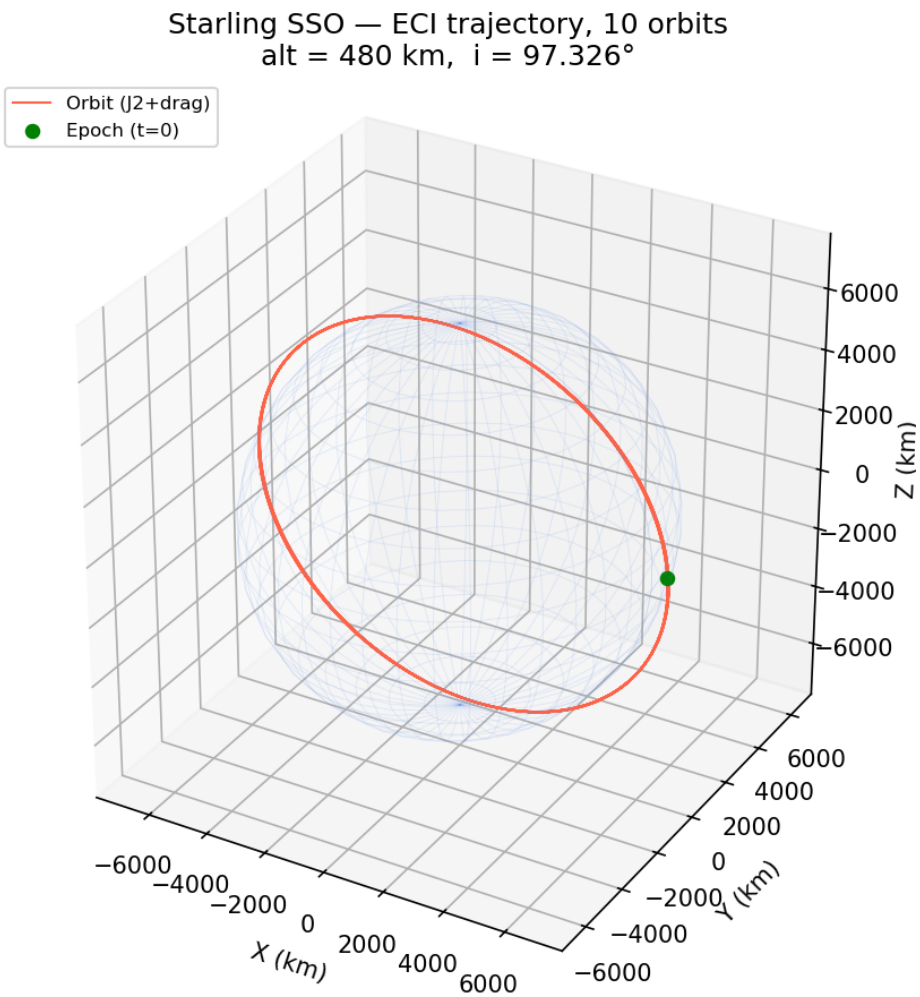

Figure 2: Simulated ECI trajectory over 10 orbits (J2 + drag). Earth wireframe shown for scale. Altitude 480 km, $i = 97.326°$.

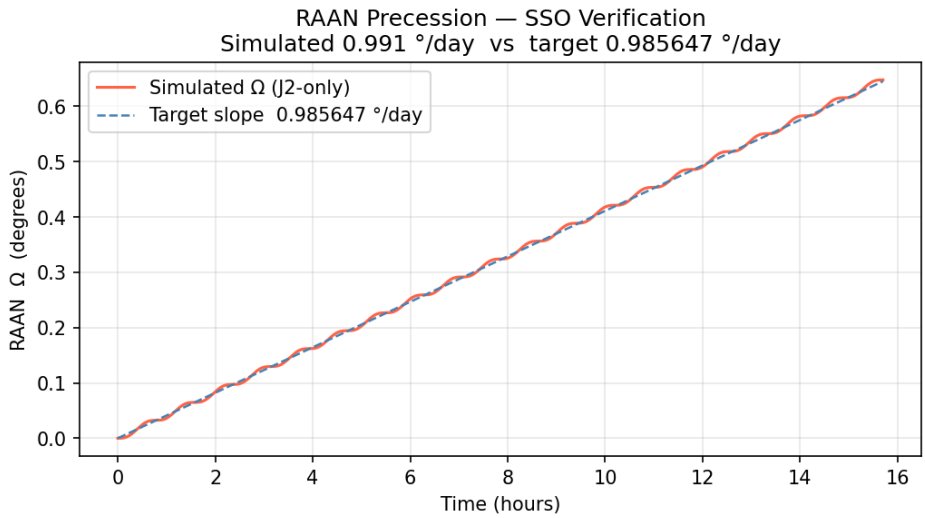

Figure 3: RAAN evolution over 10 orbits (J2-only case). Simulated precession rate 0.991 °/day versus SSO target 0.985647 °/day.

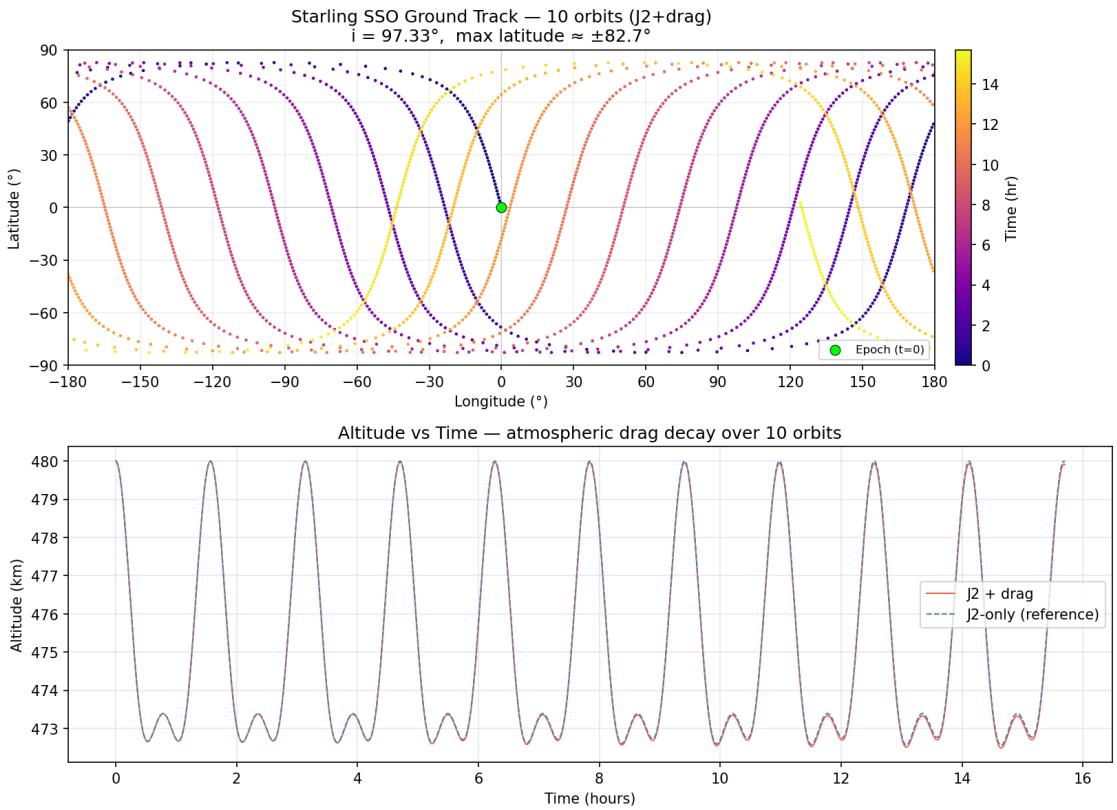

Figure 4: Top: ground track over 10 orbits (J2+drag), colored by elapsed time. The retrograde inclination (97.326°) limits ground coverage to ±82.7° latitude; each successive pass shifts ≈ 25° westward as Earth rotates beneath the satellite. Bottom: altitude vs. time showing drag-induced decay; the J2-only reference (dashed) isolates the 0.119 km SMA reduction due to drag alone.

5 Euler's Equation

5.1 Torque-Free Dynamics

In the principal frame, Euler's torque-free equations are

$$I_1\dot{\omega}_1 = (I_2 - I_3)\,\omega_2\,\omega_3, \quad I_2\dot{\omega}_2 = (I_3 - I_1)\,\omega_3\,\omega_1, \quad I_3\dot{\omega}_3 = (I_1 - I_2)\,\omega_1\,\omega_2.$$

Fix notation, simulate in body frame

These conserve two integrals of motion: the kinetic energy $T = \frac{1}{2}\boldsymbol{\omega}^\top \mathbf{I}\boldsymbol{\omega}$ and the inertial angular momentum magnitude $\|\mathbf{H}\|$. A 7-state ODE couples Euler's equations with quaternion kinematics $\dot{\mathbf{q}} = \frac{1}{2}\,G(\mathbf{q})\,\boldsymbol{\omega}$, integrated from $\mathbf{q}_0 = [1, 0, 0, 0]^\top$ using the Vern9 solver (reltol $= 10^{-10}$, abstol $= 10^{-12}$) over 600 s. The principal moments from Section 3.3 are used here.

5.2 Stability Analysis

Linearizing about a pure spin $\omega_k = \Omega$ on principal axis k, the off-axis perturbations oscillate at

$$\omega_n = |\Omega| \sqrt{\frac{(I_k - I_i)(I_k - I_j)}{I_i\, I_j}}.$$

This frequency is real and the axis is stable only when I_k is the largest or smallest principal moment. Spin along our intermediate axis yields an imaginary ω_n which leads to the tennis-racket effect described by tennis-racket theorem, which states that any perturbation will lead to unbounded tumbling.

Three simulations were run with $\|\boldsymbol{\omega}_0\| = 10$ RPM using the same angular momentum magnitude $H_0 = I_3\, \omega_{\text{mag}}$:

- **Major axis** ($I_3 = 0.249$ kg m^2, $\omega_3 = 1.047$ rad/s): 2 RPM off-axis perturbation: stable.
- **Intermediate axis** ($I_2 = 0.195$ kg m^2, $\omega_2 = 1.342$ rad/s): 0.05 RPM perturbation: unstable, separatrix.
- **Minor axis** ($I_1 = 0.140$ kg m^2, $\omega_1 = 1.868$ rad/s): 2 RPM off-axis perturbation: stable.

5.3 Nutation Period Validation

For the two stable axes the analytical nutation period is compared against the period extracted from zero-crossings of the simulated perturbed component:

Axis	**Spin (rad/s)**	**Theory T_n (s)**	**Simulated T_n (s)**	**Error**
Major (I_3)	1.047	12.942	13.019	0.59%
Minor (I_1)	1.868	9.489	9.469	0.21%

The sub-1% errors are consistent with the expected nonlinear corrections at the 20% perturbation amplitude used. The linearized formula is exact only for infinitesimal perturbations. The kinetic energy and inertial angular momentum are conserved to $|\Delta T/T_0| < 9 \times 10^{-12}\%$ and $|\Delta H/H_0| < 5 \times 10^{-12}\%$, confirming the numerical precision. Figure 5 shows these quantities vs. time for the intermediate-axis simulation.

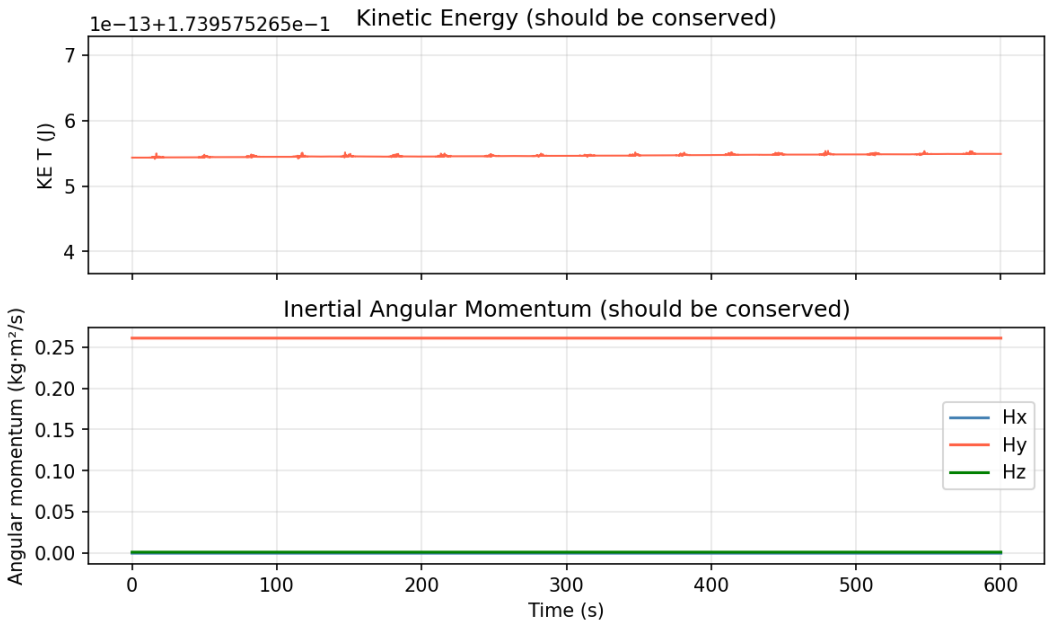

Figure 5: Conservation verification for intermediate-axis spin (torque-free, Vern9, reltol = 10^{-10}). Top: kinetic energy $T(t)$ — constant to $< 9 \times 10^{-12}\%$. Bottom: inertial angular momentum components H_x, H_y, H_z — each constant to $< 5 \times 10^{-12}\%$

5.4 Results

Figure 6 shows the angular momentum sphere in the body principal frame. As shown in class, the major and minor-axis trajectories form small closed elliptic loops around the stable equilibria ($\pm\hat{e}_3$ and $\pm\hat{e}_1$), while the intermediate-axis trajectory traces the separatrix connecting the unstable equilibria ($\pm\hat{e}_2$).

Figure 7 shows the angular velocity time histories. For stable axes the off-axis components oscillate with bounded amplitude while the intermediate axis shows large excursions in all three components.

Figure 8 demonstrates that intermediate-axis instability is independent of perturbation magnitude: perturbations of 1%, 5%, and 20% all produce the same qualitative divergence.

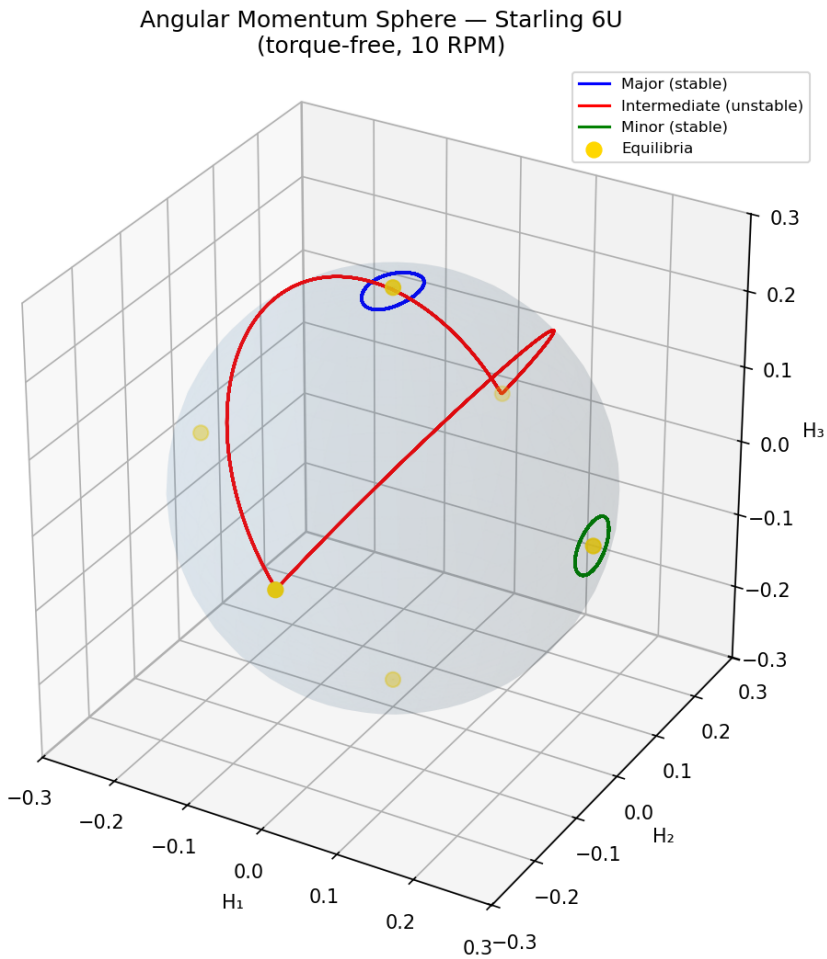

Figure 6: Angular momentum sphere for Starling 6U in the principal body frame (torque-free, 10 RPM). Blue: major-axis spin (stable, small closed loop near $+\hat{e}_3$). Green: minor-axis spin (stable, small closed loop near $+\hat{e}_1$). Red: intermediate-axis spin (unstable, separatrix). Gold dots: the six equilibrium points $\pm H_0\,\hat{e}_k$.

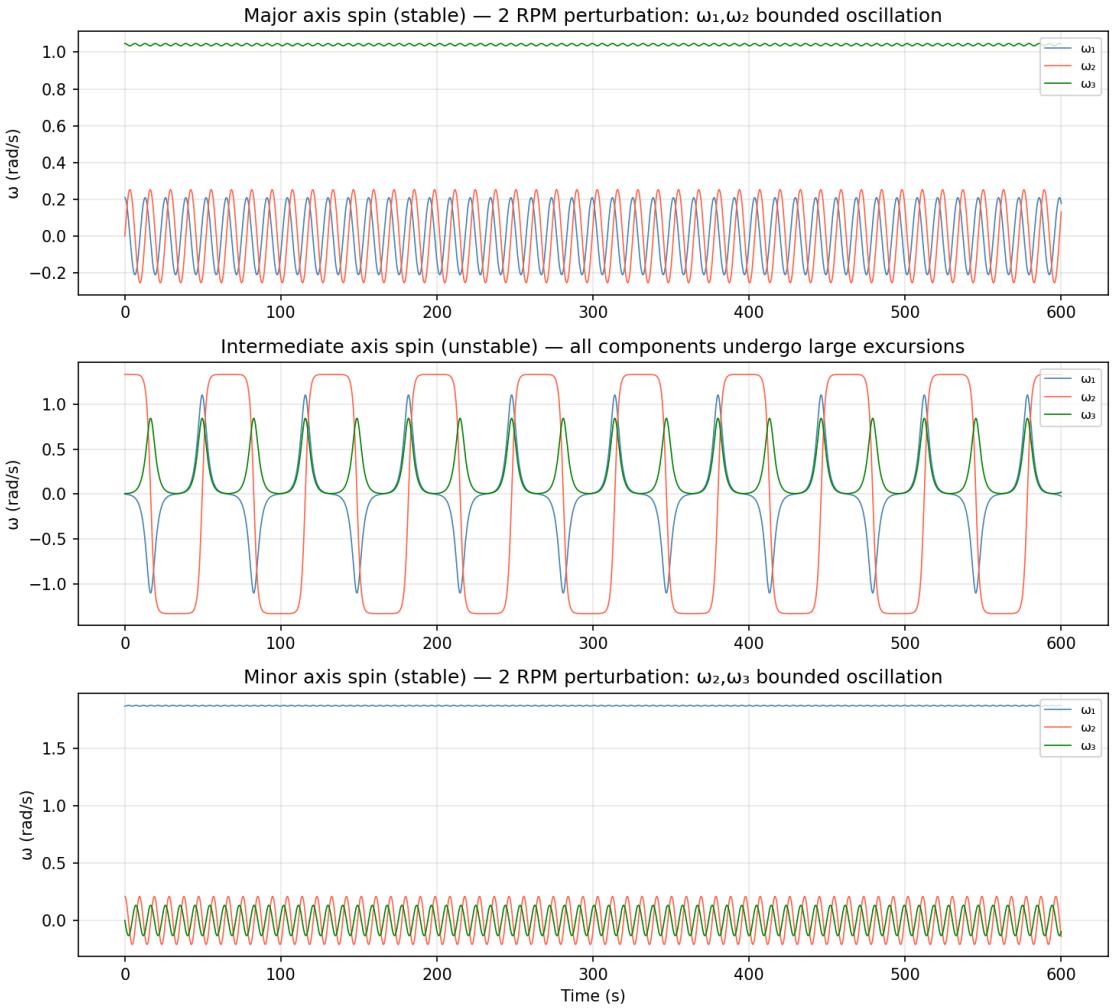

Figure 7: Angular velocity components $\omega_1, \omega_2, \omega_3$ vs. time. Top: major-axis spin — ω_1, ω_2 exhibit bounded nutation oscillations. Middle: intermediate-axis spin — all components undergo large separatrix excursions. Bottom: minor-axis spin — ω_2, ω_3 exhibit bounded nutation oscillations.

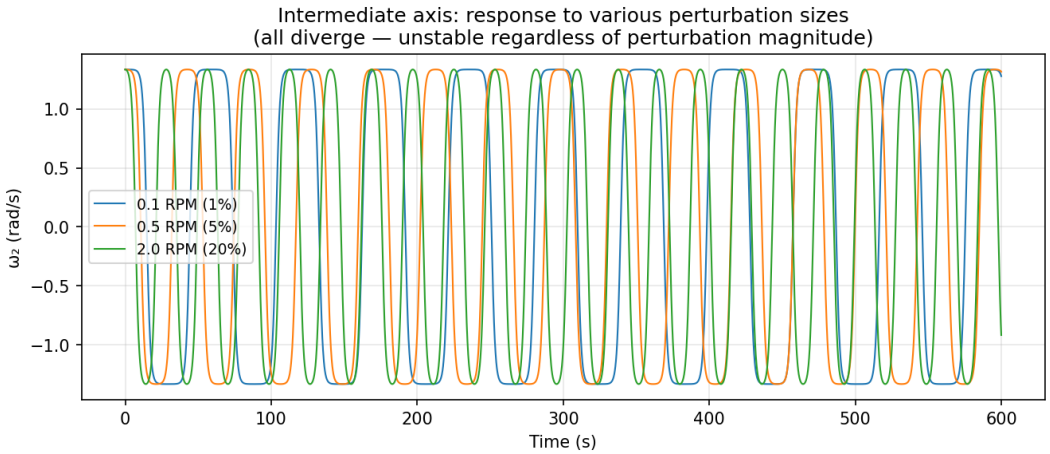

Figure 8: Intermediate-axis spin component $\omega_2(t)$ for perturbation sizes of 1%, 5%, and 20% of the spin rate. All three diverge in the same qualitative manner regardless of initial perturbation magnitude, demonstrating global instability of the intermediate axis.

6 Safe Mode Spin Design

6.1 Safe-Mode Objective

For a rigid body with three distinct principal moments of inertia, torque-free rotation is stable only about the axes corresponding to the largest and smallest principal inertias; rotation about the intermediate axis is unstable. A dual-spin or gyrostat configuration avoids this limitation by adding internal rotor momentum so that the effective inertia about the desired spin axis exceeds the transverse inertias.

For the Starling-based 6U CubeSat introduced in the previous report, the deployed solar arrays are mounted on the $\pm\mathbf{b}_1$ faces, so the solar panel normal is aligned with the body $\mathbf{b}_1$ axis when not deployed. In the body-axis ordering used previously, this corresponds to the positive body y-direction. The objective of the safe-mode design is therefore to maintain a stable spin about the panel-normal direction so that the panels remain approximately pointed toward the Sun.

6.2 Perturbed Inertia Model

The nominal inertia matrix from the CAD-based HW1 model is aligned with the spacecraft geometry, which makes the solar panel normal close to a principal direction. To avoid an artificial special case in which the desired spin axis is exactly aligned with a principal axis, the inertia matrix was randomly perturbed following the prescription in the assignment. Let the nominal inertia be decomposed as

$$\mathbf{I} = \mathbf{V}\mathbf{D}\mathbf{V}^T.$$

The principal moments were then perturbed according to

$$\widetilde{\mathbf{D}} = \mathbf{D}\left(\mathbf{I}_3 + \mathrm{diag}(\mathbf{d})\right), \qquad \mathbf{d} \sim \mathcal{N}\left(\mathbf{0}, \sigma_{\mathrm{eig}}^2 \mathbf{I}_3\right),$$

with $\sigma_{\text{eig}} = 0.03$. The principal-axis directions were perturbed by a small random rotation,

$$\widetilde{\mathbf{V}} = \mathbf{V}\exp(\widehat{\mathbf{v}}), \qquad \mathbf{v} \sim \mathcal{N}\big(\mathbf{0}, \sigma_{\text{rot}}^2 \mathbf{I}_3\big)\,,$$

where $\sigma_{\text{rot}} = 2^\circ$. The perturbed inertia matrix was then reconstructed as

$$\widetilde{\mathbf{I}} = \widetilde{\mathbf{V}}\widetilde{\mathbf{D}}\widetilde{\mathbf{V}}^T.$$

This perturbation ensures that the solar panel normal is not exactly a principal direction while preserving a physically realistic inertia tensor.

6.3 Rotor Momentum Sizing

The desired safe-mode spin rate was selected as

$$\omega_{\text{des}} = 10 \text{ RPM} = \frac{10(2\pi)}{60} \approx 1.047 \text{ rad/s},$$

about the unit vector $\mathbf{e}$ aligned with the solar panel normal. Because this axis is not, in general, a principal axis of $\widetilde{\mathbf{I}}$, passive stability is obtained by adding internal rotor momentum along the same direction.

Let the nominal inertia about the desired spin axis be

$$J_{s,0} = \mathbf{e}^T\widetilde{\mathbf{I}}\mathbf{e},$$

and let J_1 and J_2 denote the inertias about two orthonormal directions transverse to $\mathbf{e}$. Defining

$$J_t = \max(J_1, J_2),$$

the design requirement was chosen to satisfy an inertia ratio of at least

$$\frac{J_{s,\text{eff}}}{J_t} \geq 1.2,$$

where $J_{s,\text{eff}} = J_{s,0} + I_r$ includes the added rotor contribution. The required rotor inertia is therefore

$$I_r = \max\left(0,\ 1.2\,J_t - J_{s,0}\right),$$

and the corresponding rotor angular momentum is

$$H_r = I_r\,\omega_{\text{des}}.$$

For the perturbed Starling inertia used in the simulation, this calculation yielded

$$H_r \approx 0.103 \text{ N m s}, \qquad I_r \approx 0.099 \text{ kg m}^2.$$

These values indicate that a high amount of stored rotor momentum (103 mNms in the 1.0 - 100 mNms range) is required to create a gyroscopically stiff safe-mode spin about the panel normal.

6.4 Gyrostat Attitude Simulation

To verify stability, the rigid-body attitude dynamics were simulated using the torque-free gyrostat equations. The body angular velocity $\boldsymbol{\omega}$ and attitude quaternion $\mathbf{q}$ evolve according to

$$\dot{\mathbf{q}} = \frac{1}{2}\,\Omega(\boldsymbol{\omega})\,\mathbf{q},$$

and

$$\widetilde{\mathbf{I}}\,\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \left(\widetilde{\mathbf{I}}\boldsymbol{\omega} + \mathbf{h}_r\right), \qquad \mathbf{h}_r = H_r\mathbf{e}.$$

The initial angular velocity was chosen as the nominal 10 RPM spin plus a 1% transverse perturbation in order to demonstrate robustness of the configuration. The quaternion was initialized as the identity rotation for this standalone attitude-only test.

Figure 9 shows the simulated angular velocity components. The dominant spin component remains close to the commanded value, while the transverse components remain bounded and oscillatory. This is the expected response for a passively stable gyrostat: small perturbations lead to nutation, but not to runaway growth associated with intermediate-axis instability.

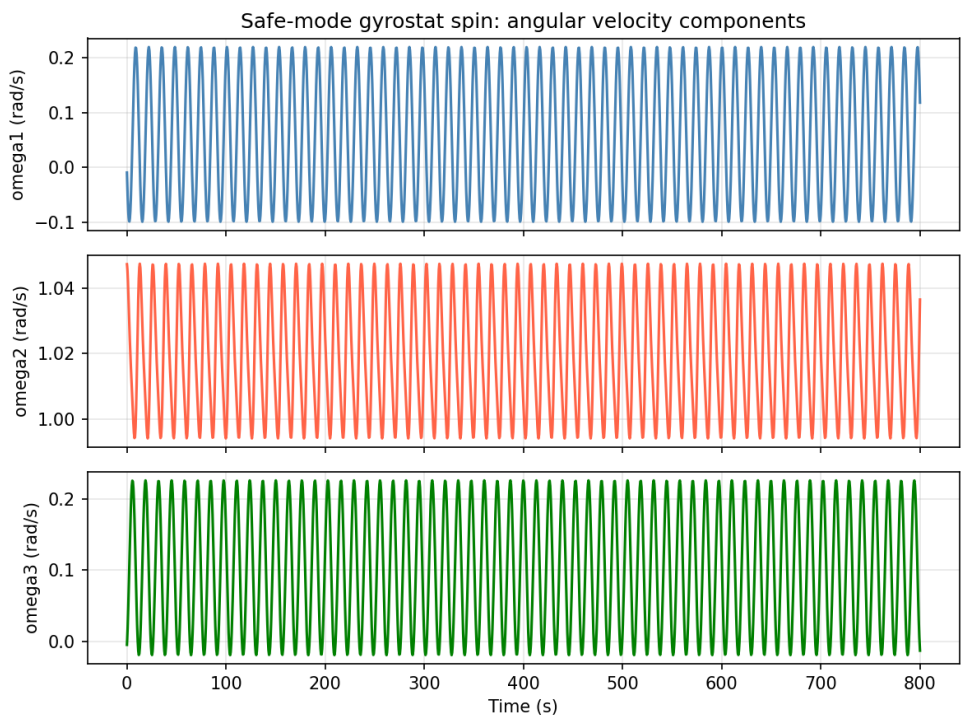

Figure 9: Safe-mode gyrostat simulation showing bounded angular velocity components under a 1% perturbation of the initial spin state.

7 Spacecraft Dynamics with Gyrostat

7.1 Coupled Orbit-Attitude Model

The second part of the assignment required extending the HW1 orbital simulation to include gyrostat attitude dynamics. The full state vector was taken as

$$\mathbf{x} = \begin{bmatrix} \mathbf{r} \\ \mathbf{v} \\ \mathbf{q} \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{R}^{13},$$

where $\mathbf{r}$ and $\mathbf{v}$ are the spacecraft position and velocity in the Earth-centered inertial (ECI) frame, $\mathbf{q}$ is the attitude quaternion mapping body coordinates to inertial coordinates, and $\boldsymbol{\omega}$ is the body-frame angular velocity.

The translational dynamics were taken directly from the HW1 model, including two-body gravity and the J_2 perturbation:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_{J_2},$$

where

$$\mathbf{a}_{J_2} = \frac{3J_2\mu R_E^2}{2r^5} \begin{bmatrix} x\,(5z^2/r^2 - 1) \\ y\,(5z^2/r^2 - 1) \\ z\,(5z^2/r^2 - 3) \end{bmatrix}.$$

Atmospheric drag was omitted for this simulation so that the effect of the coupled gyrostat dynamics could be isolated.

The rotational dynamics were again modeled with the gyrostat equation

$$\widetilde{\mathbf{I}}\,\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \left(\widetilde{\mathbf{I}}\boldsymbol{\omega} + \mathbf{h}_r\right),$$

together with the quaternion kinematics

$$\dot{\mathbf{q}} = \frac{1}{2}\,\Omega(\boldsymbol{\omega})\,\mathbf{q}.$$

7.2 Initial Conditions

The orbit was initialized as the same 480 km circular Sun-synchronous orbit used in HW1. The inclination was computed from the nodal precession requirement for a Sun-synchronous orbit, while the remaining orbital elements were selected so that the ascending node lay along the inertial x-axis. The Sun vector was assumed to be fixed along the inertial $+x$ direction, as specified in the problem.

The initial attitude quaternion was selected so that the body-fixed solar panel normal aligned with the inertial Sun vector at the start of the simulation. The initial angular velocity was again chosen as the nominal safe-mode spin plus a 1% transverse perturbation. This setup is consistent with a spacecraft that has already entered safe mode and is allowed to evolve freely under its passive gyrostat dynamics.

7.3 Simulation Results

The coupled orbit-attitude equations were integrated for 2000 s, corresponding to several nutation periods. To evaluate Sun-pointing performance, the instantaneous pointing error was defined as the angle between the inertial Sun vector $\mathbf{s}$ and the inertial image of the body-fixed panel normal,

$$\theta_{\text{err}} = \cos^{-1}\left(\frac{(\mathbf{R}_{BI}\mathbf{e}_{\text{panel}}) \cdot \mathbf{s}}{\|\mathbf{R}_{BI}\mathbf{e}_{\text{panel}}\| \, \|\mathbf{s}\|}\right),$$

where $\mathbf{R}_{BI}$ is the body-to-inertial direction cosine matrix derived from the quaternion.

Figure 10 shows the quaternion components together with the pointing error over the simulation window. The quaternion components oscillate in a bounded manner, reflecting nutation about the nominal spin axis. More importantly, the pointing error remains bounded, with no secular drift or growth in the response. The simulated behavior is consistent with a passively stable safe-mode configuration that maintains approximate Sun pointing despite perturbations in the initial spin state.

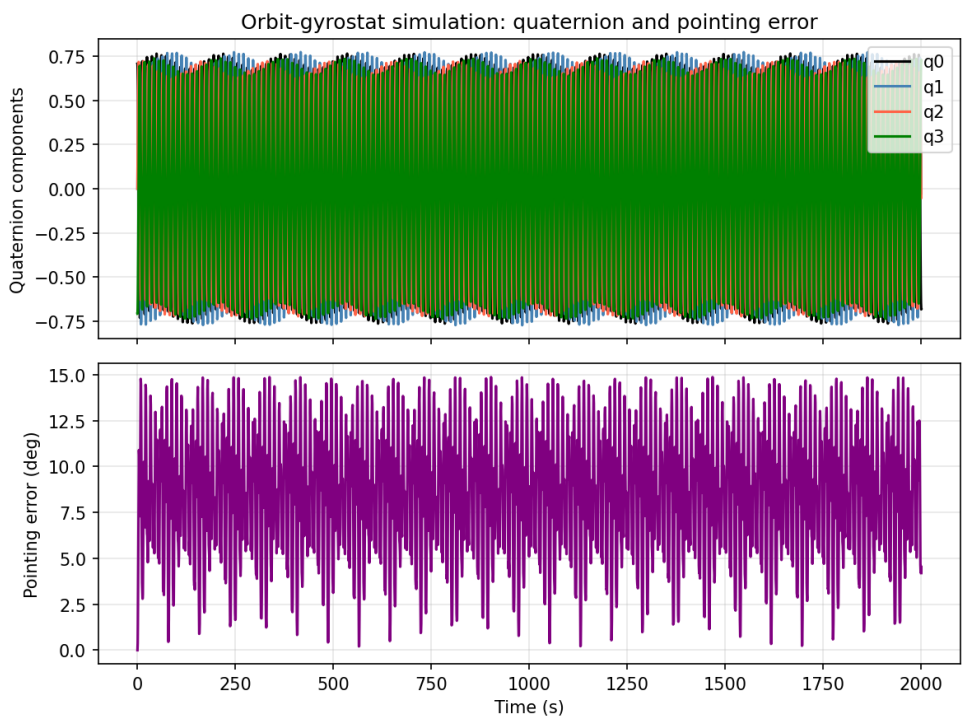

Figure 10: Coupled orbit-gyrostat simulation showing quaternion evolution and solar panel pointing error over several nutation periods.

7.4 Discussion

The safe-mode design developed here shows that a stable spin can be achieved even when the desired spin axis is not a principal axis of the nominal spacecraft inertia. By adding rotor momentum along the solar panel normal, the effective spin-axis inertia is increased

above the transverse inertias, creating the gyroscopic stiffness needed for passive stability. The resulting stored momentum is on the high side for a 6U (1-50 kg range) spacecraft, but is still within the appropriate range and is therefore compatible with a practical reaction-wheel-based implementation. [2]

The coupled orbit-attitude simulation further shows that the passive gyrostat design remains well-behaved when embedded in the full spacecraft dynamics model. The spacecraft exhibits bounded nutation, while the solar panel normal remains pointed within a limited angular envelope about the Sun direction. In an operational flight system, this safe mode would likely be entered through an active detumble and reconfiguration sequence using magnetorquers and rate feedback. Nevertheless, the results of this homework demonstrate that the passive equilibrium itself is stable and suitable as a survivable fallback attitude state for the Starling spacecraft.

8 Attitude Sensors

Published information on the mission does not appear to include a complete list of exact flight sensor part numbers, so the approach taken here is to use a Starling-consistent sensor suite based on the Blue Canyon 6U bus and standard CubeSat ADCS architecture. For this report, the modeled attitude sensor suite consists of a sun sensor, a three-axis magnetometer, and a star tracker [7, 9, 10].

8.1 Summary of Modeled Sensors

The modeled sensor suite consists of:

- a sun sensor measuring the Sun direction in the body frame, with representative performance of $< 0.3°$ accuracy and $110°$ field of view,
- a three-axis magnetometer measuring the geomagnetic field direction in the body frame, with representative performance of $\pm 60{,}000$ nT range, ≤ 8 nT resolution, and < 16 nT rms/$\sqrt{\text{Hz}}$ noise density,
- and a star tracker providing a fine attitude estimate with representative accuracy of 6 arcsec cross-boresight and 40 arcsec about-boresight.

The sun sensor and magnetometer will be used as the primary vector measurements for the static attitude estimation problem. The star tracker will also be assigned a covariance matrix and a noise model, even though it outputs attitude directly, so that its uncertainty is represented consistently within the overall sensor model.

8.2 Sun Sensor

The sun sensor provides a measurement of the Sun direction expressed in the spacecraft body frame. This makes it a natural sensor for attitude determination, safe mode recovery, and any

attitude determination problem in which the inertial Sun direction is known from ephemeris data. In this report, the sun sensor measurement is treated as a noisy unit direction vector,

$$^{B}\hat{s},$$

representing the measured Sun direction in the body frame. The AAC Clyde Space SS200 reports a field of view of 110°, an accuracy of less than 0.3° (1-σ) over the ±45° range, and a sampling rate up to 10 Hz [17].

8.3 Magnetometer

The three-axis magnetometer provides a measurement of the local geomagnetic field vector in the spacecraft body frame,

$$^{B}\hat{m}.$$

Magnetometer performance is reported in magnetic field units rather than directly as an angular error. For the representative sensor used here, the NewSpace Systems three-axis magnetometer reports a measurement range of ±60,000 nT, a resolution of at most 8 nT, a noise density below 16 nT rms/$\sqrt{\text{Hz}}$ at 1 Hz, and axis orthogonality error below 1° [18]. These values are still useful, but they require an additional modeling step before they can be used in an attitude estimation covariance matrix.

8.4 Star Tracker

The star tracker serves as the fine attitude sensor in this modeled sensor suite. Unlike the sun sensor and magnetometer, which measure individual reference directions, the star tracker estimates the spacecraft attitude directly. Blue Canyon material for nano star trackers reports attitude errors on the order of arcseconds [19].

For the sensor used here, the Blue Canyon standard nano star tracker reports a cross-boresight accuracy of 6 arcsec and an about-boresight accuracy of 40 arcsec [19]. Even though the star tracker is not used in exactly the same way as the sun sensor and magnetometer in the Wahba formulation, it should still be included in the sensor characterization. It has finite error, and that error should still be captured with a covariance model.

8.5 Sensor Covariance Matrices

8.5.1 Sun Sensor Covariance

For the sun sensor, a representative 1σ angular accuracy of 0.3° is used. Converting to radians gives

$$\sigma_s = 0.3\frac{\pi}{180} \approx 5.236 \times 10^{-3} \text{ rad}.$$

The corresponding variance is

$$\sigma_s^2 \approx 2.74 \times 10^{-5} \text{ rad}^2.$$

The sun sensor covariance matrix is therefore modeled as

$$R_s = \begin{bmatrix} 2.74 \times 10^{-5} & 0 & 0 \\ 0 & 2.74 \times 10^{-5} & 0 \\ 0 & 0 & 2.74 \times 10^{-5} \end{bmatrix}.$$

8.5.2 Magnetometer Covariance

For the magnetometer, an effective 1σ directional uncertainty of 1.0° is adopted for the attitude estimation model. Converting to radians gives

$$\sigma_m = 1.0\frac{\pi}{180} \approx 1.745 \times 10^{-2} \text{ rad}.$$

The corresponding variance is

$$\sigma_m^2 \approx 3.05 \times 10^{-4} \text{ rad}^2.$$

The magnetometer covariance matrix is then taken as

$$R_m = \begin{bmatrix} 3.05 \times 10^{-4} & 0 & 0 \\ 0 & 3.05 \times 10^{-4} & 0 \\ 0 & 0 & 3.05 \times 10^{-4} \end{bmatrix}.$$

8.5.3 Star Tracker Covariance

For the star tracker, Blue Canyon reports a cross-boresight accuracy of 6 arcsec and an about-boresight accuracy of 40 arcsec. Converting from arcseconds to radians gives

$$\sigma_\perp = 6\left(\frac{1}{3600}\right)\frac{\pi}{180} \approx 2.91 \times 10^{-5} \text{ rad},$$

$$\sigma_\parallel = 40\left(\frac{1}{3600}\right)\frac{\pi}{180} \approx 1.94 \times 10^{-4} \text{ rad}.$$

Squaring these values gives

$$\sigma_\perp^2 \approx 8.46 \times 10^{-10} \text{ rad}^2, \qquad \sigma_\parallel^2 \approx 3.76 \times 10^{-8} \text{ rad}^2.$$

Assuming the boresight is aligned with the third sensor axis, the star tracker covariance matrix is modeled as

$$R_{st} = \begin{bmatrix} 8.46 \times 10^{-10} & 0 & 0 \\ 0 & 8.46 \times 10^{-10} & 0 \\ 0 & 0 & 3.76 \times 10^{-8} \end{bmatrix}.$$

The sun sensor and magnetometer covariances are used directly in the vector measurement generation, while the star tracker covariance is used to model uncertainty in the fine attitude estimate.

8.6 Simulated Measurement Generation

With the sensor covariance matrices, simulated measurements can now be generated for use in the attitude estimation algorithms. For the sun sensor and magnetometer, the underlying

quantity of interest is a unit direction vector in the spacecraft body frame. Let the true body-frame reference vector be denoted by

$$^{B}v_{\text{true}} \in \mathbb{R}^3, \qquad \|^{B}v_{\text{true}}\| = 1.$$

A noisy sensor measurement is generated by adding zero-mean Gaussian noise with covariance R:

$$n \sim \mathcal{N}(0, R),$$
$$^{B}\tilde{v} = {}^{B}v_{\text{true}} + n.$$

Since the sun sensor and magnetometer are modeled as direction measurements, the resulting vector is then normalized so that the simulated measurement remains a unit vector:

$$^{B}v_{\text{meas}} = \frac{^{B}\tilde{v}}{\|^{B}\tilde{v}\|}.$$

$$^{B}s_{\text{meas}} = \frac{^{B}s_{\text{true}} + n_s}{\|^{B}s_{\text{true}} + n_s\|}, \qquad n_s \sim \mathcal{N}(0, R_s),$$
$$^{B}m_{\text{meas}} = \frac{^{B}m_{\text{true}} + n_m}{\|^{B}m_{\text{true}} + n_m\|}, \qquad n_m \sim \mathcal{N}(0, R_m).$$

The star tracker is modeled slightly differently. Rather than measuring a single reference vector, it provides an attitude estimate directly. Its noise is therefore represented as a small attitude error applied to the true attitude. Let the true spacecraft attitude be represented by a quaternion

$$q_{\text{true}}.$$

A small star tracker error is modeled by sampling an attitude perturbation vector

$$\delta\phi \sim \mathcal{N}(0, R_{st}),$$

where R_{st} is the star tracker covariance matrix defined previously. For small angles, this perturbation can be converted into an error quaternion

$$\delta q \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\phi \end{bmatrix},$$

followed by normalization if needed. The measured star tracker attitude is then formed by composing the true attitude with this small perturbation:

$$q_{\text{meas}} = q_{\text{true}} \otimes \delta q.$$

8.7 Monte Carlo Verification

To verify that the simulated measurements reproduce the intended noise statistics, Monte Carlo trials were run for each sensor with $N = 10{,}000$ independent samples drawn from a fixed true reference direction or attitude.

For the sun sensor and magnetometer, the true reference vector was held fixed and N noisy unit-vector measurements were generated. The angular error per sample was computed as the arc-angle between the noisy and true vectors. Because the additive Gaussian noise is isotropic in three dimensions but the result is projected onto the unit sphere, the transverse component of the noise—which drives the angular error—has two degrees of freedom. The angular error therefore follows a Rayleigh distribution with parameter σ:

$$f(\theta;\, \sigma) = \frac{\theta}{\sigma^2} \exp\left(-\frac{\theta^2}{2\sigma^2}\right), \qquad \theta \geq 0,$$

whose mean is $\sigma\sqrt{\pi/2}$ and whose RMS is $\sigma\sqrt{2}$. Table 2 summarizes the sample statistics against these theoretical predictions.

Sensor	**Intended σ**	**Theory mean**	**Sample mean**	**Sample RMS**
Sun sensor	0.300°	0.376°	0.379°	0.428°
Magnetometer	1.000°	1.253°	1.247°	1.409°

Table 2: Monte Carlo angular error statistics for vector sensors ($N = 10{,}000$). Theory and sample values agree to within 1%, confirming correct noise generation.

For the star tracker, the identity attitude was used as the true reference and N noisy quaternion measurements were generated. The perturbation vector $\delta\phi$ was recovered from each measurement as $\delta\hat{\phi} = 2\,\mathrm{logq}(q_{\mathrm{meas}})$, and the sample covariance of these vectors was compared to R_{st}. The recovered per-axis standard deviations were 5.91 arcsec, 6.00 arcsec, and 40.24 arcsec against intended values of 6, 6, and 40 arcsec, confirming the anisotropic noise model is correctly implemented.

Figure 11 shows the angular error histograms for the sun sensor and magnetometer overlaid with the theoretical Rayleigh distributions, and the per-axis perturbation histograms for the star tracker.

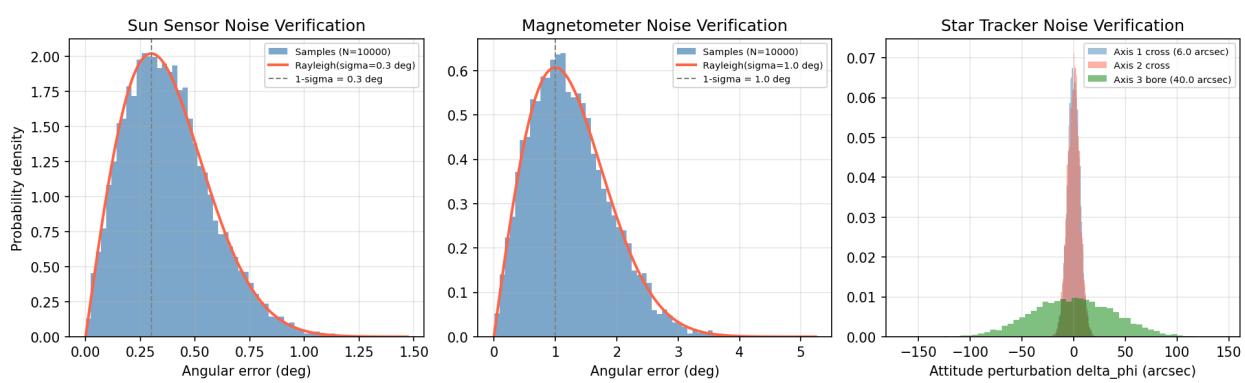

Figure 11: Monte Carlo verification of sensor noise models ($N = 10{,}000$ trials). *Left, center*: angular error histograms for the sun sensor (0.3° 1σ) and magnetometer (1.0° 1σ) overlaid with the theoretical Rayleigh distribution. *Right*: per-axis attitude perturbation histograms for the star tracker, showing the asymmetry between the 6 arcsec cross-boresight axes and the 40 arcsec about-boresight axis.

9 Static Attitude Estimation (Wahba's Problem)

9.1 Problem Formulation

Using the noisy sensor models from Section 3, we perform static attitude estimation by solving Wahba's problem [3]: given unit-vector measurements ${}^{B}b_k$ in the body frame and corresponding inertial reference vectors ${}^{N}r_k$, find the rotation matrix Q minimizing the weighted sum of squared residuals,

$$\min_{Q \in SO(3)} \sum_n w_n \left({}^{N}r_n - Q\,{}^{B}v_n\right)^T \left({}^{N}r_n - Q\,{}^{B}v_n\right),$$

where $w_k > 0$ are weights reflecting each sensor's accuracy.

9.2 Sensor Selection for Wahba's Problem

The sun sensor and magnetometer each produce a body-frame unit-vector measurement that can be paired with a known inertial reference—the Sun direction from ephemeris and the geomagnetic field from a magnetic model, respectively—making both natural inputs to Wahba's problem.

The star tracker computes a full three-axis attitude quaternion internally and its output is not a body/inertial vector pair, so it cannot enter the Wahba formulation directly here. Its role in the sensor architecture is as a standalone fine-attitude source when available.

9.3 Reference Geometry and Weights

The inertial Sun direction was taken as the ECI $+x$ axis and the magnetic reference as ${}^{N}r_m = \text{normalize}([0.3,\ 0.5,\ 0.8]^T)$. The two vectors are separated by 72 (dot product 0.303), well away from the degenerate parallel case that would leave the spin-axis degree of freedom unobservable.

Sensor weights were set proportional to inverse variance and normalized to unity:

$$w_s = \frac{1/\sigma_s^2}{1/\sigma_s^2 + 1/\sigma_m^2} \approx 0.917, \qquad w_m = \frac{1/\sigma_m^2}{1/\sigma_s^2 + 1/\sigma_m^2} \approx 0.083,$$

giving the sun sensor approximately eleven times the weight of the magnetometer.

9.4 Algorithm 1: SVD Method

The SVD solution [4] forms the attitude profile matrix

$$B = \sum_n w_n\, {}^{B}r_n \left({}^{N}r_n\right)^T,$$

computes its SVD $B = U\Sigma V^T$, and recovers the optimal rotation as

$$Q_{\text{SVD}} = V\,\text{diag}(1,\ 1,\ \det(V)\det(U))\,U^T,$$

where the diagonal correction enforces $\det(Q_{\text{SVD}}) = +1$.

9.5 Algorithm 2: Davenport q-Method

The Davenport q-method [5] works directly in quaternion space. The 4×4 Davenport matrix is assembled as

$$D = \sum_n w_n \, L^T(H\,{}^N r_n)\, R(H\,{}^B r_n),$$

where $L(\cdot)$ and $R(\cdot)$ are quaternion left- and right-multiplication matrices and $H = [0_{1\times 3};\ I_3]$ embeds a three-vector as a pure quaternion. The optimal quaternion is the eigenvector of D at its maximum eigenvalue.

9.6 Monte Carlo Comparison

Both algorithms were evaluated over $N = 5{,}000$ Monte Carlo trials with randomly generated true attitudes and noisy sun and magnetometer measurements drawn from the Section 3 models. The attitude error per trial is the three-dimensional rotation angle

$$\theta_{\text{err}} = \frac{180}{\pi} \left\| \operatorname{unhat}\!\big(\log(Q_{\text{est}}^T Q_{\text{true}})\big) \right\|.$$

Results are summarized in Table 3 and shown in Figure 12.

Algorithm	Mean	Std	RMS	95th pctile	Time (5000 trials)
SVD	0.971°	0.578°	1.130°	2.093°	0.036 s
Davenport q	0.975°	0.582°	1.135°	2.109°	0.179 s

Table 3: Monte Carlo attitude estimation accuracy and computational cost ($N = 5{,}000$ trials). Both methods solve the same optimization problem and produce statistically identical results. The SVD method is approximately five times faster.

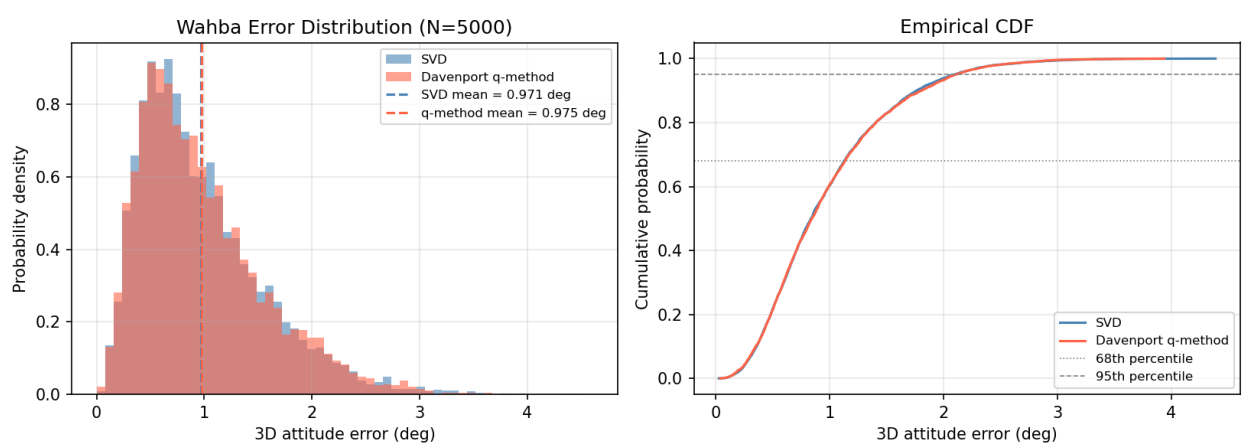

Figure 12: Monte Carlo comparison of SVD and Davenport q-method attitude estimation ($N = 5{,}000$ trials). *Left*: overlaid error histograms showing the right-skewed distribution of a 3D rotation error with sensor noise. *Right*: CDFs; the two curves overlap, confirming accuracy.

9.7 Discussion

Both algorithms are exact solvers for the same problem and produce statistically indistinguishable results. The SVD method is approximately five times faster.

The $\sim 1^\circ$ mean error is consistent with what the geometry predicts. The sun sensor constrains the two degrees of freedom transverse to the Sun direction at $\sigma_s = 0.3$; the remaining degree of freedom (rotation about the Sun vector) is constrained only by the magnetometer, whose effective accuracy along that axis is $\sigma_m / \sin\theta_{sm} \approx 1.0/\sin(72) \approx 1.05$. The expected RMS is therefore

$$\text{RMS} \approx \sqrt{2\sigma_s^2 + (\sigma_m/\sin\theta_{sm})^2} = \sqrt{2(0.3)^2 + (1.05)^2} \approx 1.13,$$

in close agreement with the simulated value of 1.13. The magnetometer is the dominant error source; The star tracker, when available, reduces this limitation by providing direct attitude measurements with arcsecond-level accuracy.

10 Attitude Sensors Revisited

This section extends the previous sensor survey by assigning concrete error structures to each sensor class: an affine distortion model for vector/bearing sensors, a multiplicative quaternion noise model for the star tracker, and an affine model with random-walk bias for the gyroscope.

10.1 Affine Vector Sensor Models

A vector sensor nominally measures the unit direction of an inertial reference vector expressed in the body frame. A more realistic model includes scale-factor error, axis misalignment, a constant bias offset, and additive noise:

$$\mathbf{y} = \text{normalize}(\mathbf{M}\,\mathbf{v} + \mathbf{b} + \mathbf{w}), \qquad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{W}), \tag{1}$$

where $\mathbf{v}$ is the true body-frame direction, $\mathbf{M}$ encodes scale factor and misalignment, $\mathbf{b}$ is the deterministic bias, and $\mathbf{W}$ is the stochastic noise covariance. The normalization step ensures the output remains a unit vector, consistent with how bearing sensors report data.

Sun Sensor (AAC Clyde Space SS200)

The SS200 datasheet [17] publishes a single, combined accuracy figure of 0.3° (1-σ) over the $\pm 45^\circ$ field of view, with no separate scale-factor or misalignment specifications. The sensor is factory-calibrated before delivery [17], so systematic pointing errors are corrected at the unit level. Because the residual post-calibration misalignment is not separately characterised in the published specification, there is no datasheet-grounded basis for constructing a non-trivial $\mathbf{M}$ or $\mathbf{b}$, and both are set to their trivial values:

$$\mathbf{M}_{\text{sun}} = \mathbf{I}_3, \qquad \mathbf{b}_{\text{sun}} = \mathbf{0}. \tag{2}$$

The noise covariance is set from the published lumped accuracy figure:

$$\sigma_{\text{sun}} = 0.3^\circ, \qquad \mathbf{W}_{\text{sun}} = \sigma_{\text{sun}}^2\,\mathbf{I}_3. \tag{3}$$

Magnetometer

The magnetometer spec sheet reports a measurement range of $\pm 60{,}000\,\mathrm{nT}$, a resolution of $< 8\,\mathrm{nT}$, a noise density of $< 16\,\mathrm{nT\,rms}/\sqrt{\mathrm{Hz}}$, and axis orthogonality below 1° before calibration [18]. The unit ships with a factory-supplied calibration matrix [18]; however, since the post-calibration residual misalignment is not separately specified, the pre-calibration orthogonality figure is used as a conservative bound on the simulation truth model. The misalignment matrix is constructed following the same convention as $\mathbf{M}_g$ in Section 10.3, using $\sin(1^\circ) \approx 1.745 \times 10^{-2}$ for the off-diagonal terms:

$$\mathbf{M}_{\mathrm{mag}} = \begin{bmatrix} 1 & 1.745 \times 10^{-2} & 0 \\ -1.745 \times 10^{-2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

Stray magnetic fields from onboard electronics produce a DC offset not characterised by the sensor datasheet. Flight measurements across CubeSat platforms show biases ranging from hundreds to tens of thousands of nT [12], with 6U platforms toward the lower end. A representative pre-calibration offset of 200 nTis adopted; dividing by the nominal LEO field strength of 45,000 nT to match the unit-vector measurement space gives:

$$\mathbf{b}_{\mathrm{mag}} = \begin{bmatrix} 4.4 \\ 2.2 \\ 3.3 \end{bmatrix} \times 10^{-3}. \tag{5}$$

This corresponds to an angular pointing error of $\sim 0.25^\circ$, below but not negligible relative to the IGRF-dominated noise budget.

Although the sensor outputs a field vector in nT, only the direction is used for attitude determination; the magnitude is discarded by normalising before the filter update. The 8 nT resolution corresponds to an angular noise floor of $\sim 0.01^\circ$ at LEO field strength, well below the IGRF model error of 0.1–0.5° [13]. The measurement noise covariance is therefore set to reflect geomagnetic model uncertainty rather than sensor noise:

$$\sigma_{\mathrm{mag}} = 1.0^\circ, \qquad \mathbf{W}_{\mathrm{mag}} = \sigma_{\mathrm{mag}}^2\, \mathbf{I}_3. \tag{6}$$

10.2 Star Tracker Model

The Blue Canyon Technologies Nano Star Tracker (NST) [19] provides a quaternion attitude measurement directly. Its noise is anisotropic: $6''$ (1-σ) in each cross-boresight axis and $40''$ (1-σ) about the boresight. A physically consistent noise model perturbs the true quaternion by a small rotation drawn from a zero-mean Gaussian:

$$\mathbf{q}_{\mathrm{meas}} = \mathrm{normalize}\big(L(\mathbf{q}_{\mathrm{true}})\, \exp_q(\boldsymbol{\phi})\big)\,, \qquad \boldsymbol{\phi} \sim \mathcal{N}(\mathbf{0}, \mathbf{W}_{\mathrm{st}}), \tag{7}$$

where $L(\mathbf{q})$ is the left-multiplication matrix and $\exp_q(\boldsymbol{\phi}) = [\cos\|\boldsymbol{\phi}\|;\ \boldsymbol{\phi}\,\mathrm{sinc}(\|\boldsymbol{\phi}\|/\pi)]$ is the quaternion exponential map. The noise covariance is diagonal in the body-frame axes aligned with the sensor:

$$\mathbf{W}_{\mathrm{st}} = \mathrm{diag}\big(\sigma_\perp^2,\ \sigma_\perp^2,\ \sigma_\parallel^2\big)\,, \qquad \sigma_\perp = 6'', \quad \sigma_\parallel = 40''. \tag{8}$$

10.3 Gyroscope Error Model

The ADIS16470 MEMS gyroscope [1] is representative of the class of small, low-power IMUs used in 6U CubeSats. Its published error budget at 10 Hz sampling consists of:

Parameter	Datasheet value	Simulation value
ARW	$0.34\ \mathrm{deg}/\sqrt{\mathrm{hr}}$	$\sigma_\omega = \mathrm{ARW}/\sqrt{h} \approx 3.13 \times 10^{-4}$ rad/s
In-run bias stability	8 deg/hr	$\sigma_\beta = (8\,\mathrm{deg/hr})\sqrt{h/3600} \approx 2.04 \times 10^{-7}$ rad/s/step
Scale factor error	$\pm 0.25\%$	$\pm 0.25\%$ diagonal entries in $\mathbf{M}_g$
Axis misalignment	$\pm 0.1^\circ$	$\sin(0.1^\circ) \approx 1.745 \times 10^{-3}$ off-diagonal in $\mathbf{M}_g$

The measurement model at time step k is

$$\tilde{\boldsymbol{\omega}}_k = \mathbf{M}_g\,\boldsymbol{\omega}_k + \boldsymbol{\beta}_k + \mathbf{v}_k, \qquad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{V}_\omega), \tag{9}$$

where the bias $\boldsymbol{\beta}_k$ evolves as a discrete random walk:

$$\boldsymbol{\beta}_{k+1} = \boldsymbol{\beta}_k + \boldsymbol{\eta}_k, \qquad \boldsymbol{\eta}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{V}_\beta). \tag{10}$$

The noise covariances are

$$\mathbf{V}_\omega = \sigma_\omega^2\,\mathbf{I}_3, \qquad \mathbf{V}_\beta = \sigma_\beta^2\,\mathbf{I}_3, \tag{11}$$

and the scale/misalignment matrix is

$$\mathbf{M}_g = \begin{bmatrix} 1.0025 & 1.745\times10^{-3} & 0 \\ -1.745\times10^{-3} & 0.9975 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{12}$$

Figure 13 shows the bias random walk simulated over one hour (36 000 steps at 10 Hz). All three axes remain within the $\pm 1\sigma$ envelope set by the 8 deg/hr in-run bias stability specification.

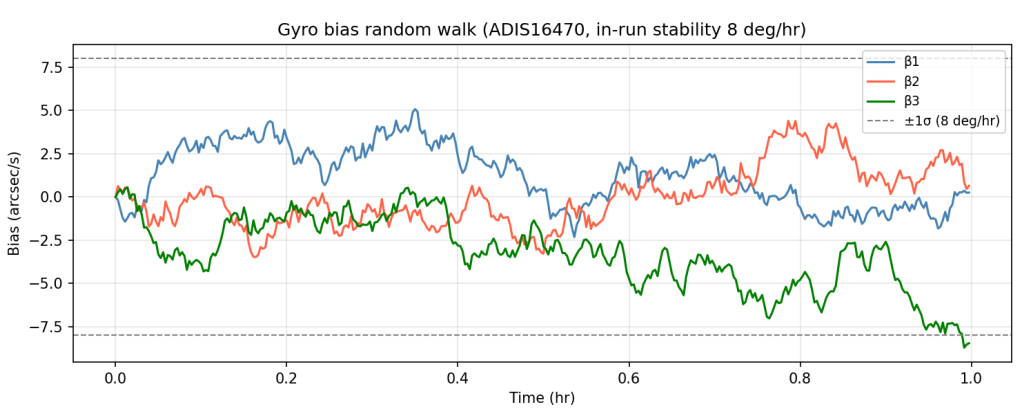

Figure 13: Simulated ADIS16470 gyroscope bias random walk over one hour at 10 Hz. Dashed lines show the $\pm 1\sigma$ (8 deg/hr) in-run bias stability envelope.

11 Recursive Attitude Estimation

11.1 Sampling Rate

Our sampling rate for attitude estimation system was selected based on both the spacecraft rotational dynamics and its sensors' capabilities. The Blue Canyon XACT on board support slew rates on the order of 3-5 deg/s for a 6U craft, which is taken as an upper bound for an uncontrolled tumble.

For a worst-case tumble rate of 5 deg/s, the rotation period is about 72 s. The dominant dynamic mode is the nutation motion, which for our inertia properties has a period in the order of 10-15 s. To resolves this, a minimum of 10 samples per nutation period was selected, allowing a max allowable timestep of approximately 1 s. This satisfies this requirement and provides sufficient (?) resolution for accurate state estimations. To assess the impact of a different sampling rate and its improvement on performance, we also conducted tests at 0.5 s and 1 s.

11.2 Filter vs. Static Error

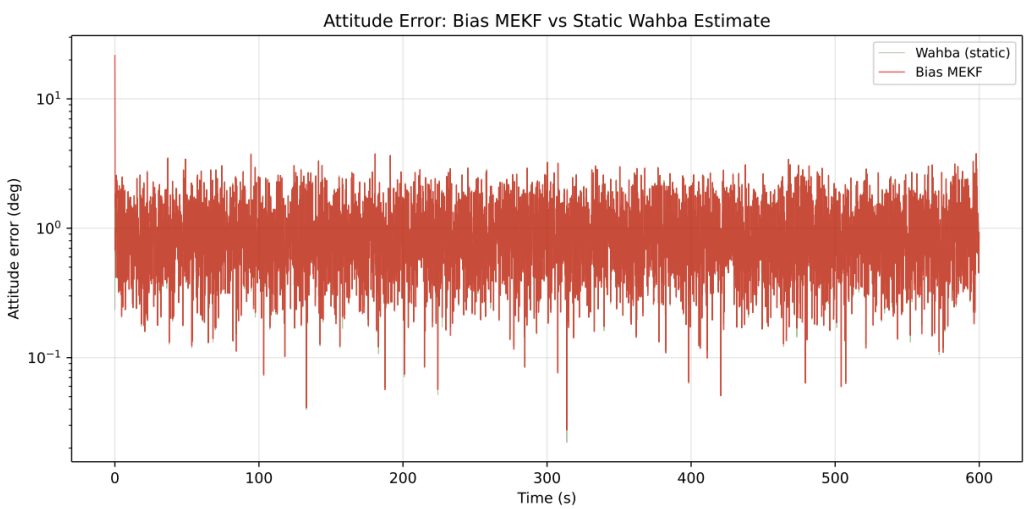

Figure 14: Attitude Error for Bias MEKF vs. Static Wahba

The error for both the Wahba estimate and the Bias MEKF remain on the order of

$$10^1$$

degrees, being slightly offset in the negative direction. Both remain relatively similar throughout as we can observe by the overlap between the two.

11.3 Estimator Consistency

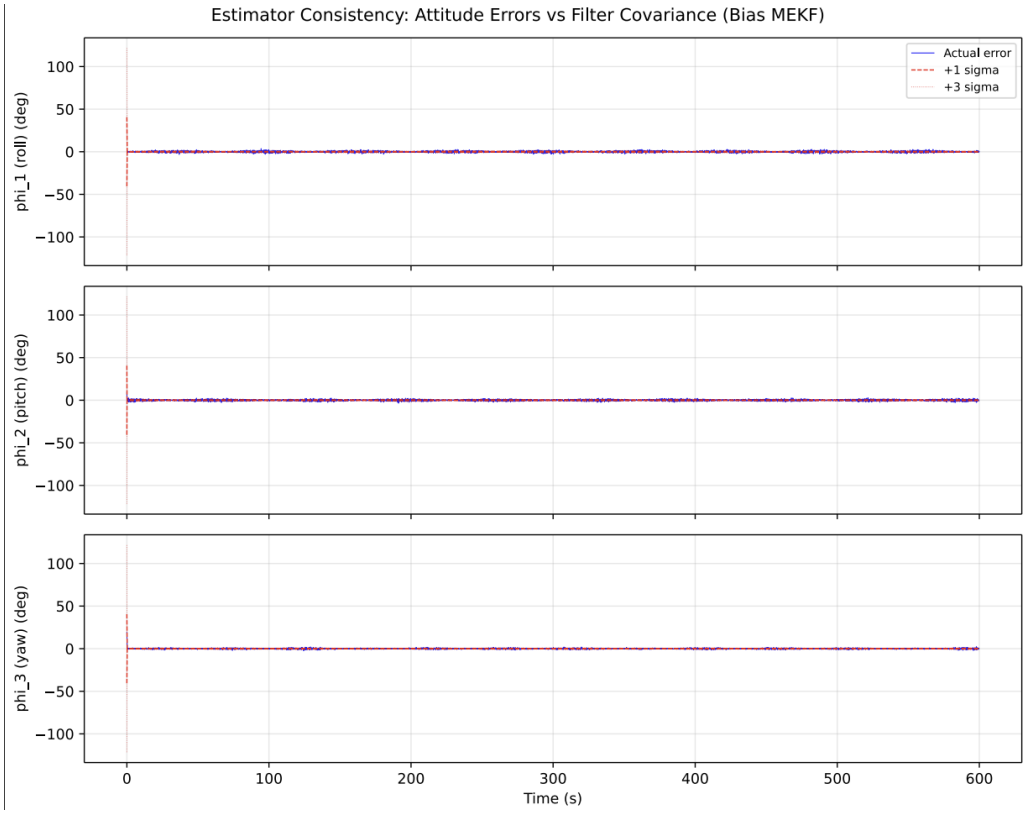

Figure 15: Attitude Erros vs Filter Covariance for Bias MEKF

We produce a bounded bias error history consistent within the covariance envelope. We can conlcude that the random walk bias model functions well and as intended, which support the use of our bias MEKF

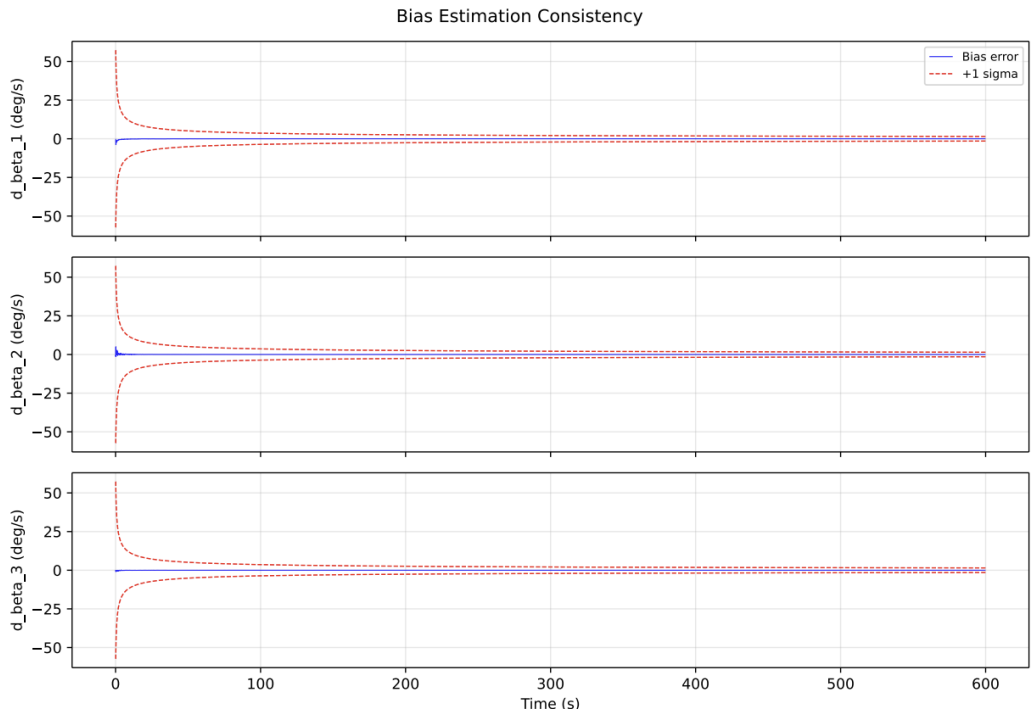

Figure 16: Bias Consistency

The bias estimation consistency also seems to stay well within reasonable bounds. The precision of both of these show that the filter is properly tuned.

11.4 Convergence Behavior at Different States

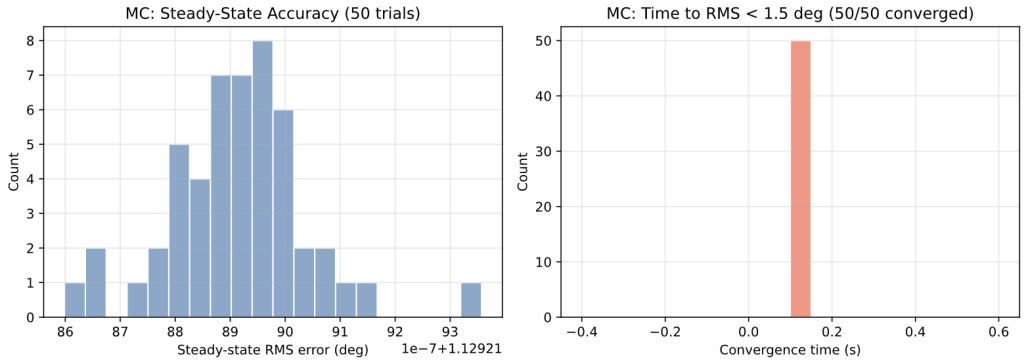

Figure 17: Steady State Accuracy and Time to ¡1 degree for 50 Trails

Our RMS very closely matches the 1.3 predicted in the previous analyzes of sensors using only bias MEKF.

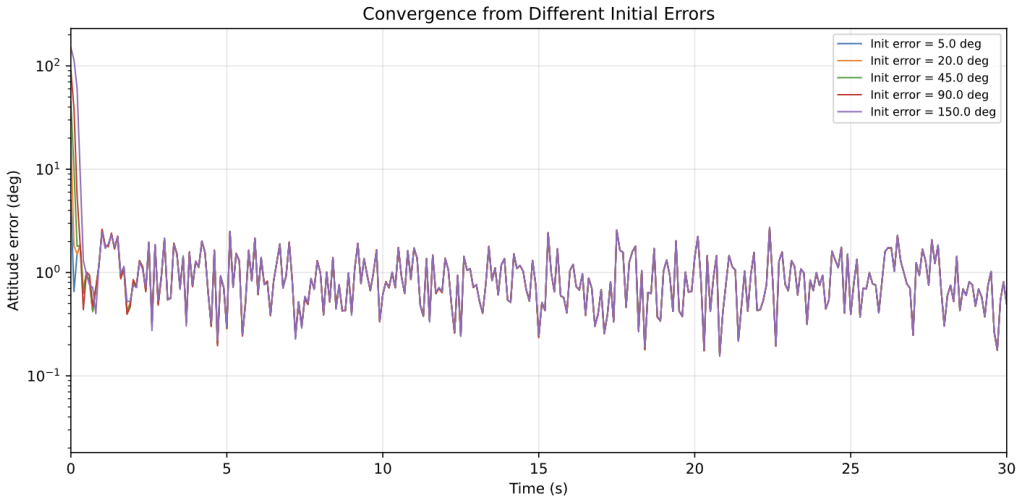

Figure 18: Convergence from different Initial Errors

As we can observe, the attitude error converges nicely, much before the 5 second mark, again staying relatively bounded. This confirms that, at least at low tumble, the bias MEKF is not heavily affected by substantial changes in initialization values, usually converging quickly to the same error band.

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